

Rishi Ratnakar

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PROFESSIONAL SUMMARY

Data analyst and analytics engineer with a B.S. in Statistical Data Science. Builds reproducible SQL and Python pipelines, statistical analyses, machine-learning models, and Power BI reporting across consumer finance, urban mobility, and product experimentation. Brings statistical-consulting exposure and a strong focus on data quality, transparent documentation, and decision-ready communication.

EDUCATION

Virginia Tech, Blacksburg, VA

2026

Bachelor of Science in Statistical Data Science

Relevant coursework: Methods of Regression Analysis, Intermediate Data Analytics and Machine Learning, Methods of Statistical Computing, Experimental Designs, Data Analytics and Visualization, Communication in Statistical Collaboration

Northern Virginia Community College

2020

Career Studies Certificate in Information Systems Technology, Database Specialist

TECHNICAL SKILLS

Languages: Python, SQL, R

Analytics: A/B testing, regression, statistical modeling, exploratory data analysis, data quality, machine learning

Data and BI: Power BI, DAX, Tableau, Excel, pandas, DuckDB, BigQuery, Databricks, PySpark, Delta Lake, dbt, Parquet

Development and training: Git, GitHub Actions, pytest, Ruff, Linux; Microsoft Fabric data engineering (DP-700 preparation)

PROJECTS

Product Experimentation and Customer Analytics

Python, SQL, DuckDB, BigQuery, Power BI, scikit-learn

- Designed a deterministic Python and DuckDB analytics pipeline for 20,000 synthetic users and 129,937 events, separating event-grain behavior from user-level assignment and outcomes to prevent double-counting.
- Produced ordered funnels, cohort retention, and segment metrics; estimated a 2.76-percentage-point treatment lift (95% CI: 1.65-3.87; $p = 1.05e-6$) and evaluated assignment balance and guardrails.
- Trained a leakage-aware retention classifier with a chronological split (ROC-AUC 0.641; average precision 0.406), using only information available before the prediction window.
- Passed 20 of 20 persisted data-quality checks and 11 automated tests in GitHub Actions; delivered four Power BI pages and separately executed a documented BigQuery funnel against Google's public GA4 sample.

Urban Mobility Analytics Engineering

Databricks, PySpark, Delta Lake, dbt, SQL, Power BI

- Built a Databricks medallion pipeline for 7.05 million official NYC taxi records, using PySpark and Delta Lake to produce 6.56 million validated Silver trips and 13,884 Gold daily-zone records.
- Implemented schema enforcement, deduplication, idempotent Delta MERGE logic, and late-arrival handling to make repeated pipeline runs reliable and auditable.
- Built dbt models and tests, passed 8 of 8 cloud data-quality checks, and automated local validation through GitHub Actions.
- Published a four-page Power BI operations report and documented the architecture, validation controls, and reproducible execution path for technical review.

Consumer Finance Complaint Intelligence

Python, CFPB API, DuckDB, SQL, Power BI, scikit-learn

- Ingested and analyzed 5,442,964 CFPB complaints from 2025 through a reproducible Python, Parquet, DuckDB, and SQL workflow.
- Designed a 5,000-record weekly-stratified detail sample to preserve time coverage for interactive reporting while keeping the full population available for aggregate analysis.
- Developed a leakage-aware relief classifier with 73.3% ROC-AUC and 48.2% average precision using a time-aware validation design.
- Added nine automated tests and a three-page Power BI report, with documented sampling methods, data-quality results, and interpretation limits.

RELEVANT EXPERIENCE

Statistical Consulting Shadowing and Professional Development

Jan-Jul 2026

Virginia Tech Statistical Applications and Innovations Group (SAIG), Blacksburg, VA

- Observed weekly statistical consulting meetings and documented how analysts translated client questions into scoped analyses, selected appropriate methods, and communicated assumptions and limitations.
- Prepared an ADEPT presentation on measurement quality, explaining accuracy, precision, repeatability, and reproducibility through practical examples for a nontechnical audience.
- Worked with a JMP-based water-quality activity designed for learners, translating statistical concepts and visual exploration into accessible instructional language.
- Synthesized consulting observations and assigned readings in written reflections focused on reproducibility, collaborative practice, and communicating statistical results responsibly.

Summer Extern

Jun-Aug 2020

United States Patent and Trademark Office, Alexandria, VA

- Supported Oracle database monitoring and enterprise technical workflows using Python and Linux, gaining exposure to operational reliability within a large federal technology environment.
- Contributed to technical documentation and presentations and used Git to collaborate on shared technical work.